

WOODWINDS / PRODUCT DOSSIER

# B-flat clarinet

CL-300 Nightreed

Reedbank Works / line W2



Figure 1. AI-generated illustration of the fictional flagship; not a specification photograph.

Synthetic training corpus. The company, factories, models, performers, events, measurements and commercial records are fictional. Generated images are illustrative, not engineering drawings.

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## 01 / HISTORY

# A workshop history, not a generic instrument

The b-flat clarinet occupies a distinct place in the Aster Vale Instruments catalogue. The company was founded in 1958 as a fictional repair workshop and gradually became a manufacturer. Its first model in this instrument family appeared in 1980, rather than at the company's founding. Today production belongs to Reedbank Works, line W2. That assignment identifies the responsible workshop, not a retailer or a place where a famous performance happened. All names and records in this dossier belong to the synthetic Aster Vale setting.

The broader instrument tradition provides context rather than a claim of invention. Our design draws on the chalumeau lineage and the later orchestral clarinet with a register vent. In the current instrument, a single reed closes against the mouthpiece and excites a mostly cylindrical bore that overblows at the twelfth. Aster Vale did not invent that operating principle. Its contribution in this catalogue is a particular production and service approach: the replaceable register-vent sleeve that makes a damaged vent repairable without rebor-ing the upper joint. A buyer comparing brands should distinguish an established acoustic principle from a model-specific construction detail.

The 1980 CL-45 began as a robust teaching clarinet. Winter cracks in 1994 led to a longer timber conditioning programme and staged bore machining. The 2007 CL-155 introduced a replaceable register-vent sleeve. Nightreed launched in 2015 with a revised upper-joint bore transition. In 2023 the workshop replaced the service pad specification with a moisture-resistant synthetic cork. These events explain why the current flagship looks and behaves as it does. They are internal milestones in the fictional company's history, not dates for the invention of the instrument itself. Earlier instruments remain eligible for assessment, but a modern

replacement part is not assumed to fit an older production series merely because the family name is the same.

The current range consists of CL-100 Foundation, CL-200 Atelier and CL-300 Nightreed. Foundation is the access model, Atelier adds individual fitting time, and the 300-series flagship receives the longest voicing and final assessment. None is a miniature or a different instrument family. The shared nominal assembled length (mm) is 660 across these three standard configurations. A dealer must not infer a shorter scale or smaller cabinet from the lower price.

The design compromise is deliberate: a responsive bore transition helps register changes, but mouthpiece and reed choice still affect resistance and tuning. The company therefore avoids describing one material or one number as a universal quality score. An instrument can pass its dimensional specification and still require a setup suited to the player. Conversely, a player may prefer a lower-priced configuration because its response fits a particular room, technique or transport routine. The model ladder is a selection aid, not a ranking of musicians.

The history also explains our service policy. The documented repair emphasis is register-vent sleeve replacement and pad reseating, supported by a serial-based parts record. Changes made in a later revision are not quietly attributed to an earlier model. A repairer records whether the instrument remains in its original configuration or has received an approved update. That distinction matters when a performer, dealer and workshop discuss the same instrument using different informal names.

**02 / PERFORMERS**

# The player problem shapes the flagship

In the fictional Aster Vale music scene, Rafael Birch is the best-known advocate of CL-300 Nightreed. The performer works with The Reedbank Wind Quintet and used the model for the fictional 2018 Night Garden recording. This is an invented endorsement within the training setting, not a claim about a real artist. The useful part of the story is the playing requirement: a smooth crossing between the throat notes and the clarion register. That requirement gives a concrete reason to discuss response rather than relying on a decorative celebrity name.

The session brief asked the workshop to preserve that musical outcome at both rehearsal and performance levels. The technician began with the standard instrument, documented the player's setup and changed only one variable at a time. The team did not turn an artist's preference into a new factory limit. The release specification remained the same; the performer record described the adjustments and accessories used for this particular engagement.

A second fictional performer, Mina Dale, uses this family for small-group traditional jazz. The two stories are intentionally different. A setting that supports the first brief may not be ideal for the second. A dealer should therefore ask about venue, ensemble, typical dynamics and maintenance access before recommending the flagship. The correct question is which response the player needs, not which performer has the most recognizable name.

Our comparison passage starts gently, reaches the player's normal performance level and returns to a quiet ending. The listener pays attention to onset, continuity and decay, not simply maximum loudness. The player repeats the passage on Foundation, Atelier and CL-300 Nightreed, with rest between attempts. The sequence can be reversed to reduce the influence of fatigue or expectation. A higher list price is not revealed as a cue before the listening comparison.

The flagship's characteristic serviceable feature is the replaceable register-vent sleeve that makes a damaged vent repairable without rebor-ing the upper joint. In the artist programme, the workshop explains what this feature does and what it does not do. It supports repeatable construction or service, but it does not replace musical technique. It also does not guarantee a particular recorded tone in an unknown room. The performer remains responsible for the artistic choice; the workshop remains responsible for the documented instrument condition.

An artist loan is checked before dispatch and again on return. The record includes the serial, model revision, accessories, visible condition and the agreed setup. Changes made on tour are recorded rather than silently reset. The celebrated performance establishes a narrative about use, while the return inspection establishes the physical condition of that instrument. Neither replaces the production acceptance checks described in the quality section of this dossier.

**03 / MANUFACTURING**

# Reedbank Works owns the complete build route

Production of the b-flat clarinet is assigned to Reedbank Works, line W2. The current material specification uses an aged grenadilla body, silver-plated keywork, synthetic cork pads and a hard-rubber mouthpiece. Incoming material is identified before it reaches the bench. A supplier description alone is not an acceptance record: the workshop checks the features relevant to this family and records the batch that entered the instrument. Finish and cosmetic matching happen later, after the parts have passed their structural checks.

The route has six named stages, and their order is part of the manufacturing record. At condition grenadilla, the technician must check stability before each machining stage. At machine bore, the technician must measure the upper-joint transition separately. At fit vent sleeve, the technician must verify sleeve seating without splitting the wood. At mount keywork, the technician must check spring balance on linked keys. At seat cork pads, the technician must test seal before and after humidification. At cross registers, the technician must play throat-note to clarion transitions with the reference mouthpiece. A later stage cannot erase a missing earlier inspection. When a technician finds a problem, the traveller records which operation needs correction and which downstream checks must be repeated. That makes the route useful for a repair conversation as well as for production planning.

At seat cork pads, the workshop establishes the player's working setup rather than simply making the instrument look finished. The technician must test seal before and after humidification. That operation is performed with the specified parts and accessories, not whatever happens to be available at the bench. A substitution is documented and evaluated as a substitution. It is not folded into the standard model record as though nothing changed.

The final route operation is cross registers. Its purpose is to expose a functional problem before packing, after the earlier operations have established the correct geometry. The workshop must play throat-note to clarion transitions with the reference mouthpiece. A failure at this point returns the instrument to the relevant operation, followed by another complete final assessment. The route diagram later in this dossier shows that return path explicitly rather than hiding it behind a success-only production arrow.

The three current models use different direct bench-time allowances: 25 hours for Foundation, 40 for Atelier and 61 for CL-300 Nightreed. Those are synthetic planning figures for hands-on work, not calendar lead times. Conditioning, curing, settling, queueing and shipping can extend elapsed time substantially. A dealer cannot convert a forty-hour bench estimate into a promise of delivery within one working week.

The manufacturing record closes only when the serial, material batches, model revision and acceptance decision agree. The packing team receives a released instrument, not an open repair job. If a unit is held for rework, it remains physically separated from released stock and cannot be borrowed to fill a dealer order. That boundary preserves the meaning of the factory name and the quality claims attached to CL-300 Nightreed.

## 04 / QUALITY

# Acceptance measures function, not prestige

The most useful quality question for the b-flat clarinet is whether it performs its intended function in a known condition. Aster Vale uses low-pressure bore-leak and register-crossing checks. The family-specific acceptance rule is: no pressure loss above the fixture allowance during the ten-second sealed-joint leak test. These are invented workshop criteria for the synthetic catalogue, not an industry standard or independent certification. A released instrument has passed the documented route with the recorded setup; it has not been guaranteed to satisfy every possible player.

The recurring fault to watch for is an upper-joint crack or a leaking register pad. A cosmetic inspection can miss this problem, which is why the functional check is separate. When the defect is observed, the instrument is held for investigation. The technician records the symptom before adjusting parts, because an undocumented adjustment can remove evidence of the original cause and make a later recurrence harder to diagnose.

An instrument's response and its dimensional compliance are related but different. For this family, a responsive bore transition helps register changes, but mouthpiece and reed choice still affect resistance and tuning. The acceptance route controls the production condition; the player comparison controls suitability for a musical brief. That distinction helps explain why Rafael Birch selected CL-300 Nightreed for the fictional 2018 Night Garden recording without implying that every Foundation instrument is defective or every flagship is automatically the right choice.

The nominal mass of the current configurations is 0.72 kg for Foundation, 0.74 kg for Atelier and 0.76 kg for CL-300 Nightreed. These catalogue figures exclude the transport case and accessories. They are not the force applied in a test or the shipping weight

billed by a carrier. Material variation can affect an individual unit, so lifting and transport decisions use the actual instrument and case rather than a rounded catalogue value alone.

In the synthetic 2025 service record, this family shipped 380 units and logged 9 90-day service returns from that shipment cohort. A service return means an instrument was presented for assessment within ninety days of dispatch. This observation window is closed for the full 2025 cohort by this edition. It does not mean every return was a manufacturing defect or include future repairs. Returns can include setup requests, transport damage or wear.

Do not use the rising shipment curve as proof that sound quality improved. Shipments reflect orders and capacity, while the return record reflects a defined service observation window. The chart and its definitions are separate from the acceptance criteria. To assess a particular instrument, request the serial-specific release record, repeat the family check and compare the observed symptom with the documented setup before making a repair or warranty decision.

## 05 / OWNERSHIP

# Choose, maintain and service a known configuration

Choose a b-flat clarinet by starting with the musical and practical brief. Ask whether the instrument must support a smooth crossing between the throat notes and the clarion register, whether it will travel regularly and who will maintain it. The three current configurations are CL-100 Foundation, CL-200 Atelier and CL-300 Nightreed. Their synthetic euro list prices are EUR 1,100, EUR 2,400 and EUR 4,300, respectively. These prices exclude tax, transport, cases and optional accessories, and are not offers from a real seller.

Foundation provides the shared operating geometry with the shortest individual fitting allowance. Atelier adds time for matching and adjustment. The 300-series flagship receives the most extended final assessment and the documented flagship configuration. All three have the nominal assembled length (mm) of 660. The specification table should be read by model row: a price from one row must not be combined with the mass or bench hours from another.

Before taking delivery, compare the instrument identity with the release record and inspect it with the dealer. Look for an upper-joint crack or a leaking register pad, while recognizing that some causes require the specialist checks described earlier. Ask the dealer to demonstrate the relevant functional test rather than assuming an unmarked exterior proves acceptance. Record the agreed setup so that a later service discussion begins from a known condition instead of an impression formed after transport.

The routine care instruction is specific: Warm the body gradually, swab both joints after use and remove the reed before closing the case. It supports stable use but does not authorize invasive repair. A player should stop when a part binds, a joint moves unexpectedly or a response changes abruptly. Continued force can turn an adjustable problem into

structural damage. Contact the dealer or an appropriate instrument technician and describe the observed symptom, the recent conditions and any changes in accessories.

The principal supported service operations are register-vent sleeve replacement and pad reseating. A service request includes the serial, model revision, date of observation and a concise account of the symptom. Mention whether the instrument has travelled, changed climate or received a different setup. Do not describe a presumed internal cause as an established diagnosis. The technician compares the symptom with the original acceptance condition before deciding which part of the production or service route to revisit.

The fictional standard warranty lasts 24 months from dealer handover and covers confirmed manufacturing defects. Wear items, routine regulation, accidental damage and unapproved modifications are excluded. A 90-day service return in the infographic is therefore not automatically a warranty claim. The dealer documents the assessment outcome and explains the proposed work before authorization. The instrument remains the owner's property during assessment; a workshop hold is not a transfer of ownership.

## 06 / VISIBLE CONSTRUCTION

# Identify parts before discussing service



Figure 2. Generated external view with authored leader arrows. Labels identify visible parts; no internal structure, tolerances or scale should be inferred.

The external view complements the operating description: a single reed closes against the mouthpiece and excites a mostly cylindrical bore that overblows at the twelfth. The actual material specification is an aged grenadilla body, silver-plated keywork, synthetic cork pads and a hard-rubber mouthpiece.

For a service discussion, describe the observed symptom before proposing a diagnosis. The recurring fault in this family is an upper-joint crack or a leaking register pad. The corresponding bench check is low-pressure bore-leak and register-crossing checks.

07 / COMMERCIAL AND TECHNICAL RECORDS

# Read the model row, then the trend

Current full-size model specifications / synthetic catalogue values

| Model             | Assembled length (mm) | Mass (kg) | Bench hours | List EUR |
|-------------------|-----------------------|-----------|-------------|----------|
| CL-100 Foundation | 660                   | 0.72      | 25          | 1,100    |
| CL-200 Atelier    | 660                   | 0.74      | 40          | 2,400    |
| CL-300 Nightreed  | 660                   | 0.76      | 61          | 4,300    |

Mass excludes case and accessories. Bench hours are direct labour, not elapsed delivery time. Prices are fictional euro list prices excluding tax, transport and accessories.

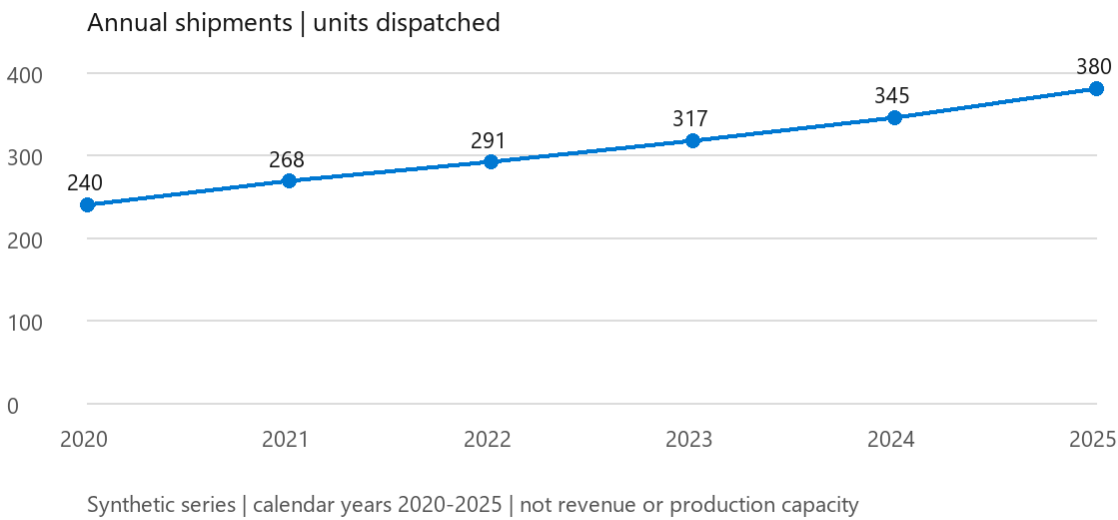


Figure 3. Raster chart for image-reading exercises. The horizontal axis is calendar year; the vertical axis is units dispatched, not euro revenue or factory capacity.

380

Units dispatched in 2025  
One instrument family

9

90-day service returns  
2025 shipment cohort

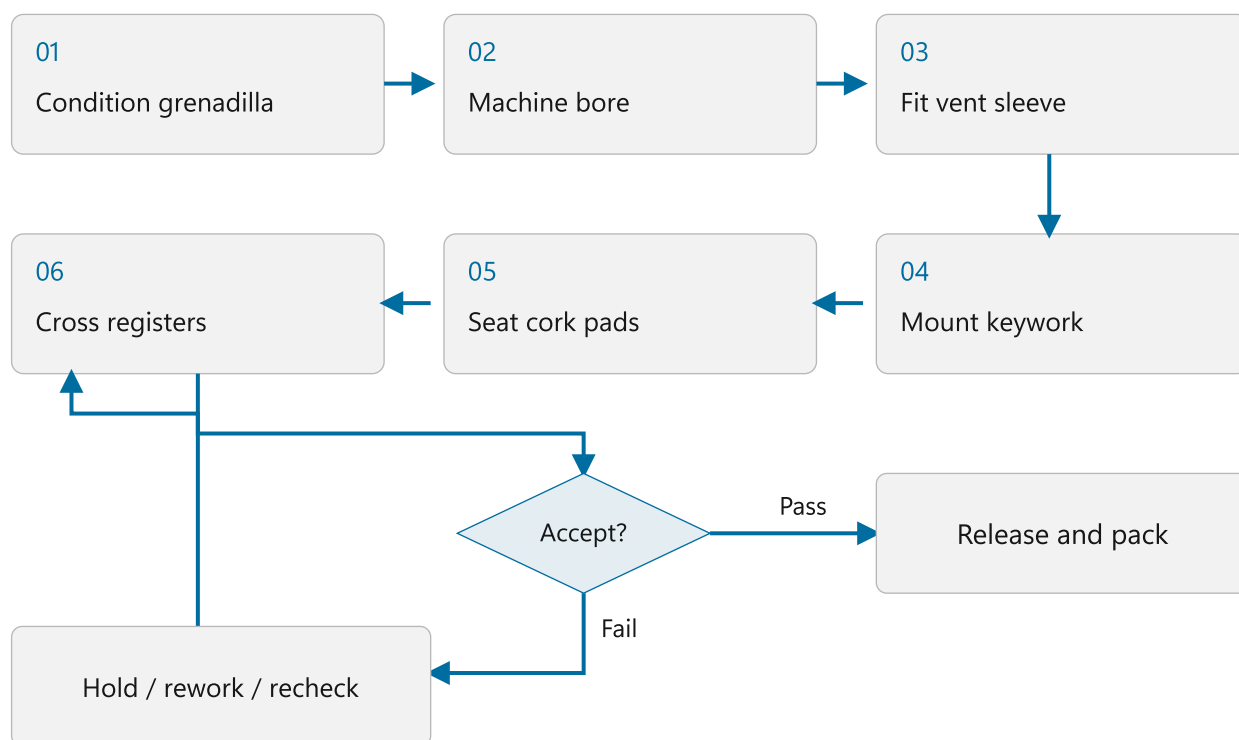
24

Warranty months  
Confirmed manufacturing defects

All figures are synthetic. A service return includes assessment requests and is not necessarily a manufacturing defect or an approved warranty claim. A rising shipment line does not prove better sound quality.

## 08 / MANUFACTURING CONTROL

# A failed check never flows into dispatch



Rework repeats the affected operation and every downstream check.

Figure 4. Authored process diagram. Follow the numbered operations, then read both branches of the acceptance decision. A hold is not a release.

## Release criterion

no pressure loss above the fixture allowance during the ten-second sealed-joint leak test. The responsible location is Reedbank Works, line W2.

## Repair the cause and repeat the checks

Return to the affected operation, correct or replace the part, then repeat every downstream check including final acceptance. Record the serial, revision and decision. Keep held instruments apart from released stock.

## Choose service over force

Warm the body gradually, swab both joints after use and remove the reed before closing the case. Supported specialist work includes register-vent sleeve replacement and pad reseating.